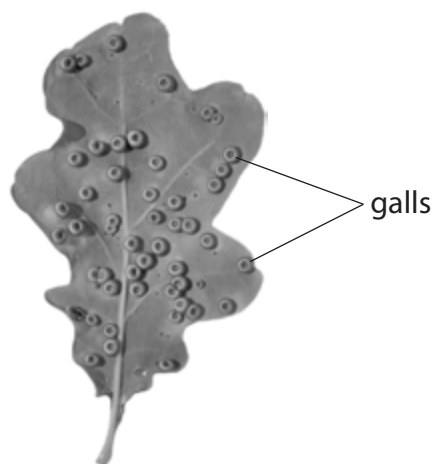


- 7 Some insects lay eggs inside leaves. This causes the leaves to produce swellings called galls.

The galls supply the developing insects with nutrients and protect them from the external environment.

The photograph shows galls on a leaf.



In an investigation, the concentrations of protein molecules and amino acids found in healthy leaves, leaves with galls and in the galls themselves were determined.

- (a) The insects were removed from each gall before the investigation.

Suggest why the insects were removed from the galls.

(2)

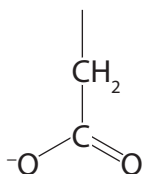


(b) The amino acids were extracted and dissolved in a non-polar organic solvent.

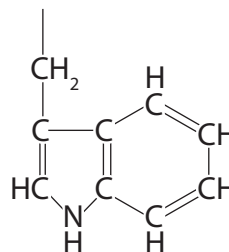
- (i) The diagram shows the R group of the amino acids, alanine, aspartate and tryptophan.



alanine



aspartate



tryptophan

Which row of the table describes the solubility of these amino acids in a non-polar organic solvent?

(1)

Solubility in a non-polar organic solvent			
	Most soluble	→	Least soluble
☐ A	alanine		aspartate
☐ B	alanine		tryptophan
☐ C	aspartate		alanine
☐ D	aspartate		tryptophan

- (ii) The solubility of an amino acid can be determined by measuring the maximum mass of the amino acid that dissolves in a known volume of solvent.

The solubility of the amino acid histidine in a solvent is 43.5 g dm^{-3} .

The mass of the amino acid leucine that dissolves in 250 cm^3 of the same solvent is 5.5 g .

Calculate how many times more soluble histidine is than leucine.

(2)

Answer

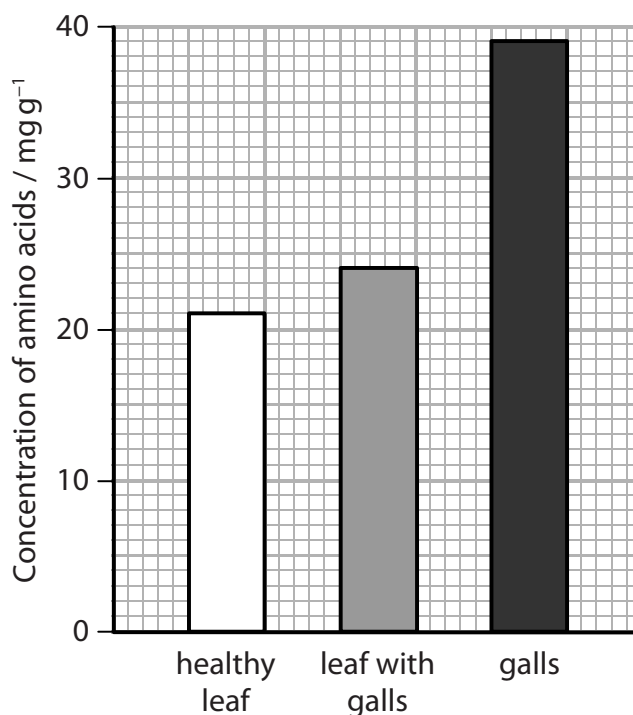


*(c) The gall-forming insects secrete saliva into the plant tissues.

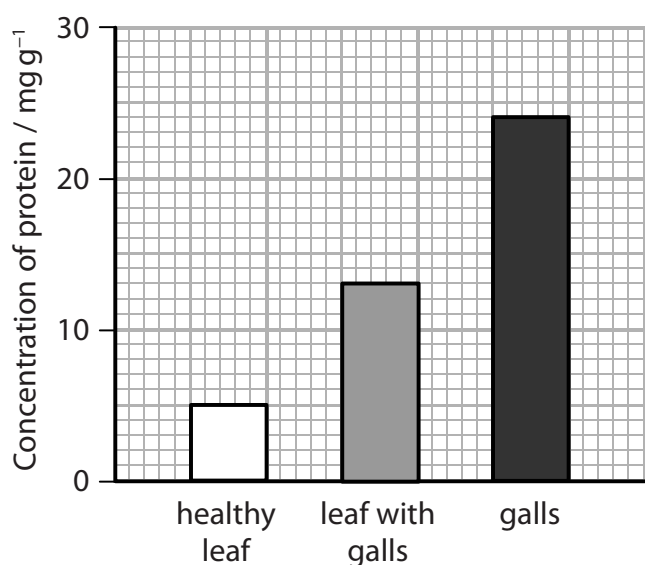
The saliva contains enzymes that change the nutrients in the leaf and cause the galls to form.

The graphs show the concentrations of protein molecules and amino acid molecules in tissues from a healthy leaf, tissues from a leaf with galls and in the galls themselves.

Amino acid molecules



Protein molecules



The table shows the abundance of five amino acid molecules.

Amino acid	Abundance in tissues of a healthy leaf	Abundance in tissues of a leaf with galls	Abundance in galls
alanine	++	+	++
arginine	++	—	++
histidine	—	—	++
leucine	+	+	+
tryptophan	—	—	—

Key

++ most abundant

— absent

Explain the results of this investigation. Use the information in both graphs and the table to support your answer.

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 7 = 11 marks)

